



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

30-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	A	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C,D	A	B,C,D	A,B,C	B,C	B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	205.00 to 210.00	4.00	16.00	15.00	2.11 to 2.13
SECTION-II (ii)	Q.	7	8	9			
	A.	25	2	2			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	D	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C,D	A,B,C	B,D	B,C,D	A,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	45.60	30.00	4.00	7.00	5.00
SECTION-II (ii)	Q.	7	8	9			
	A.	2	8	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	B	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,B,D	A,B	A,C	C,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.00	-0.25	16.00	0.00	6.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	10	0	4			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

$$kA = p$$

$$\frac{1}{2}m\omega^2 A^2 = q \Rightarrow \frac{1}{2}m\omega^2 \frac{p^2}{k^2} = q$$

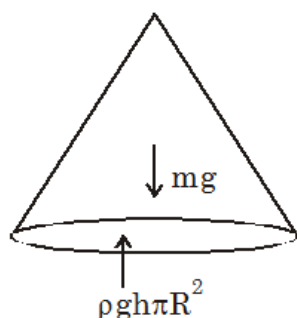
$$\frac{1}{2}m\omega^2 \frac{p^2}{m^2\omega^4} = q \Rightarrow \omega = \frac{p}{\sqrt{2mq}}$$

2. Ans (D)

$$T \rightarrow \frac{T}{2}$$

$$f \rightarrow 2f$$

3. Ans (A)



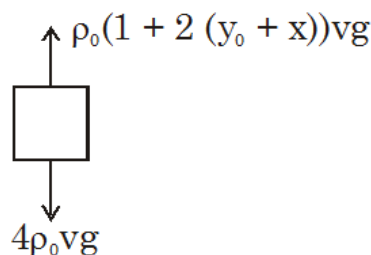
$$\rho gh\pi R^2 = mg + F_{\text{side}}$$

$$\rho gh\pi R^2 = \frac{1}{3}\pi R^2 h \rho g + F_{\text{side}}$$

$$F_{\text{side}} = \frac{2}{3}\pi R^2 h \rho g$$

4. Ans (A)

$$4\rho_0 = \rho_0 (1 + 2y_0) \Rightarrow 3 = 2y_0 \Rightarrow y_0 = \frac{3}{2}$$



$$F_{\text{net}} = -\rho_0 \left(1 + 2 \left(\frac{3}{2} + x \right) \right) vg - 4\rho_0 vg$$

$$= -\rho_0 (4 + 2x) vg - 4\rho_0 vg$$

$$ma = -\rho_0 2vgx$$

$$4\rho_0 v a = -\rho_0 2vgx$$

$$a = -\frac{gx}{2} \Rightarrow \omega = \sqrt{5}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (C,D)

displacement in one time period is 0.

$$v_{\text{avg}} = 0$$

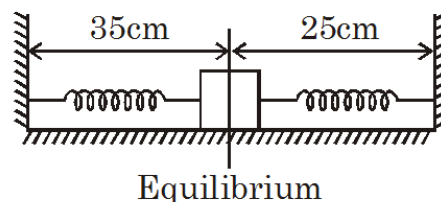
6. Ans (A)

$$k_1 x_1 = k_2 x_2 \text{ and } x_1 + x_2 = 20$$

$$x_1 = 3x_2$$

$$x_1 = 15 \text{ cm}$$

$$x_2 = 5 \text{ cm}$$



$$k_{\text{eq}} = 4000$$

$$\omega = \sqrt{\frac{4000}{100} \times 1000} = 200$$

$$T = \frac{2\pi}{200} = \frac{\pi}{100}$$

7. Ans (B,C,D)

$$v = \sqrt{a - bx^2}$$

$$v = \sqrt{b} \sqrt{\frac{a}{b} - x^2}$$

$$\omega = \sqrt{b} = 2\pi f$$

$$T.E. = \frac{1}{2} k A^2 = \frac{1}{2} m b \left(\frac{a}{b} \right) = \frac{ma}{2}$$

8. Ans (A,B,C)

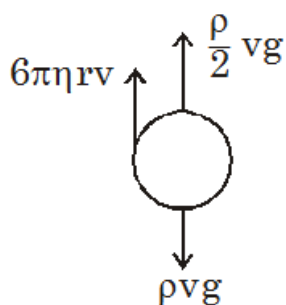
$$v = \frac{v_0}{H^2} \left(2y^2 - \frac{y^3}{H} \right)$$

$$\frac{F}{A} = \eta \frac{dv}{dy} = \eta \frac{v_0}{H^2} \left(4y - \frac{3y^2}{H} \right)$$

$$(A) \text{ at } y = \frac{H}{2} ; \frac{F}{A} = \eta \frac{v_0}{H^2} \left(2H - \frac{3H}{4} \right) = \eta \frac{5v_0}{4H}$$

$$(B) \text{ at } y = H; \frac{F}{A} = \eta \frac{v_0}{H^2} (4H - 3H) = \frac{\eta v_0}{H}$$

9. Ans (B,C)



$$\frac{\rho}{2} vg = 6\pi\eta rv$$

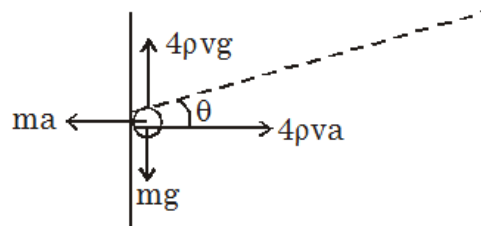
$$\frac{\rho}{2} \frac{4}{3} \pi r^3 g = 6\pi\eta rv$$

$$v = \frac{r^2 \rho g}{9\eta}$$

$$mgH + W = \frac{1}{2} \rho \frac{4}{3} \pi r^3 \left(\frac{r^2 \rho g}{9\eta} \right) - \frac{m}{2} gH$$

$$|W| = \frac{4}{3} \pi r^3 \rho \left[\frac{gH}{2} - \frac{1}{2} \left(\frac{r^2 \rho g}{9\eta} \right)^2 \right]$$

10. Ans (B,C)



$$\tan \theta = \frac{4\rho vg - mg}{4\rho va - ma}$$

$$\frac{1}{2} = \frac{g}{a}$$

$$a = 2g$$

$$\frac{1}{2} a_y t^2 = \frac{L}{2}$$

$$\frac{1}{2} 3gt^2 = \frac{L}{2} \Rightarrow t = \sqrt{\frac{L}{3g}}$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (205.00 to 210.00)

$$\frac{dV}{dt} = av = a\sqrt{2gh}$$

4. Ans (16.00)

$$ma = F = \frac{-dU}{dx} = -32(x-1)$$

$$a = -16(x-1) \Rightarrow \omega = 4 \text{ rad/second}$$

At mean position $a = 0 \Rightarrow x = 1$ (mean position)

Applying conservation of mechanical energy

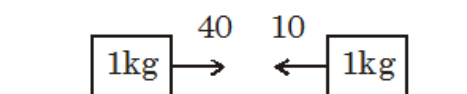
$$0 = -16 + K. \epsilon_{\max} \Rightarrow K. \epsilon_{\max} = 16 \text{ Joule}$$

6. Ans (2.11 to 2.13)

(15 to 16)

$$x = 8 + 4 \sin(10t)$$

$$k = 100 \frac{N}{m}$$



$$40 \times 1 - 10 \times 1 = 2v$$

$$30 = 2v$$

$$v = 15$$

$$v = \omega A$$

$$A = \frac{15\sqrt{2}}{10}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (25)

$$T = 2\pi \sqrt{\frac{\mu}{k}} = 2\pi \sqrt{\frac{6}{5 \times 300}} = 2\sqrt{\frac{6}{5 \times 30}}$$

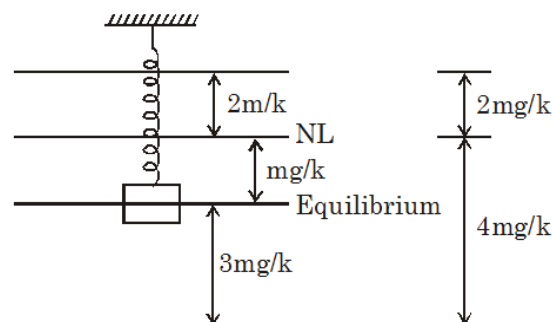
$$T = \frac{2}{5}$$

$$nT = 60$$

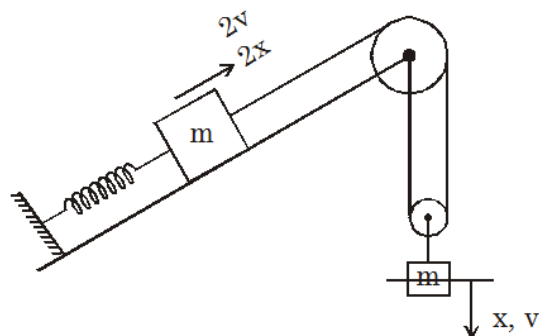
$$n \frac{2}{5} = 60 \Rightarrow n = 150 = 6N$$

$$N = 25$$

8. Ans (2)



9. Ans (2)



$$TE = \frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 + \frac{1}{2}k(2x)^2$$

$$TE = \frac{1}{2}mv^2 (5) + \frac{1}{2}k(2x)^2 \quad (\text{Using energy method})$$

$$\omega = \sqrt{\frac{4k}{5m}} = 2\sqrt{\frac{k}{5m}}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (B,D)
(I) incorrect M.O.
(II) Hund's rule violation
(III) ground state configuration
(IV) 3 e⁻ in each orbital

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (1.50)

$$\rho = \frac{2 \times \frac{450}{6 \times 10^{23}}}{(10 \times 10^{-8} \text{ cm})^3} = 1.5 \frac{\text{gm}}{\text{cc}}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (2.00)

$$\text{Density} = 8 = \frac{Z_{\text{eff}} \times M}{a^3 \times N_A}$$

$$\Rightarrow M = \frac{8 \times (200 \times 10^{-10})^3 \times 6 \times 10^{23}}{4}$$

$$= 9.6 \text{ g/mol}$$

$$\therefore \text{No. of atoms in 32 gm} = \left(\frac{32}{9.6} \times 6 \times 10^{23} \right)$$

$$= 2 \times 10^{24}$$

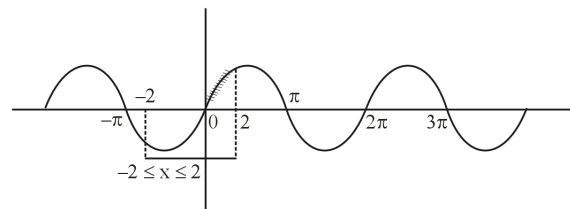
$$N = 2$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$4 - x^2 > 0; 2 < x < 2$$



for $f(x)$ to be define $x \in (0, 2]$

2. Ans (D)

$$\sin^{-1}\sin(7) + \cos^{-1}\cos(5) = (7 - 2\pi) + (2\pi - 5) = 2$$

3. Ans (B)

$$f(x) = 4x^3 - 12x = 4x(x^2 - 3)$$

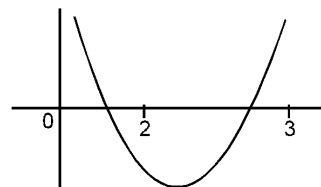
$$f(0) = f(\sqrt{3}) = f(-\sqrt{3}) = 0$$

$\Rightarrow$ Many one function

Similarly onto function as range is $[-8, 72]$

4. Ans (A)

$$x^2 - \frac{2p}{p-5}x + \frac{p-4}{p-5} = 0$$



$$f(0) > 0, f(2) < 0, f(3) > 0$$

$$f(0) > 0 \Rightarrow \frac{p-4}{p-5} > 0 \quad \dots(1)$$

$$f(2) < 0 \Rightarrow \frac{p-24}{p-5} < 0 \quad \dots(2)$$

$$f(3) > 0 \Rightarrow \frac{4p-49}{p-5} > 0 \quad \dots(3)$$

Intersection of (1) (2) & (3) gives $\left(\frac{49}{4}, 24 \right)$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C)

$$f(x) = \sqrt{1-x^2} \left(\frac{1}{1+x^2} - x^2 \cdot \frac{x^2}{1+x^2} \right)$$

$$= \sqrt{1-x^2} (1-x^2) = (1-x^2)^{3/2}$$

Range of $f(x)$ is $[0, 1]$

$$f\left(\frac{\sqrt{3}}{2}\right) = \left(1 - \frac{3}{4}\right)^{3/2} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

6. **Ans (A,B,D)**

$\therefore$ sum of all coefficient = 0 $\Rightarrow x = 1$ is a root and product of roots

$$= \frac{c + 2a - 3b}{a + 2 - b - 3c} = \frac{1}{4}$$

$\therefore$ roots are $1, \frac{1}{4}$.

7. **Ans (A,B)**

(a) $f(1) = f(2) = \dots = f(101) = 0$. function is many one.

(b) $f(x)$ is odd degree polynomial and it is onto

8. **Ans (A,C)**

$$\alpha = \tan^{-1} \left(\frac{4}{3} \right) + \tan^{-1} \left(\frac{3}{4} \right) = \frac{\pi}{2}$$

$$\beta = \tan^{-1} \left(\frac{12}{5} \right) + \tan^{-1} \left(\frac{3}{4} \right) + \tan^{-1} \frac{63}{16}$$

$$= \pi - \tan^{-1} \left(\frac{63}{16} \right) + \tan^{-1} \frac{63}{16} = \pi$$

$$\therefore 2 \sin \alpha = 2 \text{ \& } \cos \pi = -1$$

$$\therefore \text{equation } x^2 - x - 2 = 0$$

$$p = 1 \text{ \& } q = -2$$

9. **Ans (C,D)**

$$h(x) = -f(x) + g(x)$$

$$= \begin{cases} 2 & ; x \in \text{Rational} \\ 6x + 1 & ; x \in \text{Irrational} \end{cases}$$

clearly $h(x)$ is many one into

10. **Ans (A,B,D)**

$f(x)$ is periodic even and unbounded function

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (8.00)**

$$\text{Let } f(x) = ax + b$$

if f is increasing then

$$f(-1) = 3 \Rightarrow -a + b = 3$$

$$f(1) = 5 \Rightarrow a + b = 5$$

$$\text{hence } f(x) = x + 4$$

$$f(x) = x + 4$$

$$f\left(-\frac{1}{4}\right) = \frac{15}{4} \Rightarrow f^{-1}\left(\frac{15}{4}\right) = -\frac{1}{4}$$

2. **Ans (-0.25)**

$$\text{Let } f(x) = ax + b$$

if f is increasing then

$$f(-1) = 3 \Rightarrow -a + b = 3$$

$$f(1) = 5 \Rightarrow a + b = 5$$

$$\text{hence } f(x) = x + 4$$

3. **Ans (16.00)**

$$x(x-2)(x-4)(x-6) + 16$$

$$= (x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$$

$$\text{put } x = 4$$

4. **Ans (0.00)**

$$f(x) = x(x-2)(x-4)(x-6) + 16$$

$$= (x^2 - 6x)(x^2 - 6x + 8) + 16$$

$$\text{let } x^2 - 6x + 4 = t$$

$$= (t-4)(t+4) + 16$$

$$= t^2$$

5. **Ans (6.00)**

$$\sin^{-1}x \text{ is } \uparrow$$

$$\tan^{-1}x \text{ is } \uparrow$$

$$\sqrt{x+1} \text{ is } \uparrow$$

$$\Rightarrow f(x) \text{ is } \uparrow \text{ in its domain i.e. } [-1, 1]$$

$$f_{\min} = f(-1) = -\frac{3\pi}{4}$$

$$f_{\max} = f(1) = -\frac{3\pi}{4} + \sqrt{2}$$

$$\Rightarrow f(x) \in \left[-\frac{3\pi}{4}, \frac{3\pi}{4} + \sqrt{2} \right]$$

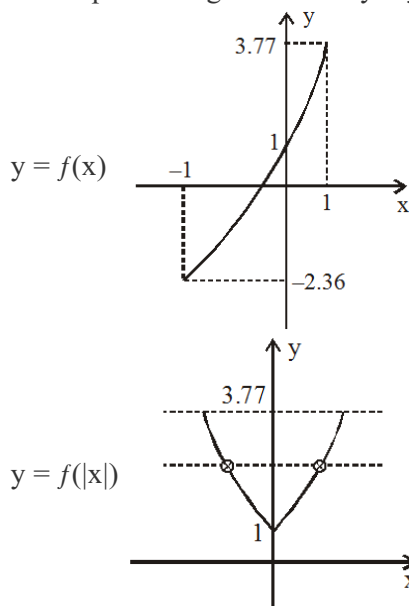
$$\Rightarrow f(x) \in [-2.36, 3.77]$$

$$\Rightarrow k = -2, -1, 0, 1, 2, 3$$

$$\Rightarrow \text{Number of values} = 6$$

6. **Ans (2.00)**

Let us plot a rough sketch of $y = f(x)$



$\Rightarrow f(|x|) = 2$ has exactly two solutions

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (10)**

Put $x = 1$, we get $f(2) = 2f(1)$

Put $x = 2$, we get $f(3) = 3f(1)$

Similarly $f(n) = nf(1)$

$$\therefore \sum_{r=1}^n f(r) = f(1) \cdot [1 + 2 + 3 + \dots + n]$$

$$\Rightarrow \frac{n(n+1)}{2} = 55 \Rightarrow n = 10$$

8. **Ans (0)**

$$x^2 + |x + 3| + |x - 3| = 24$$

$$\text{case (i) } x > 3 \Rightarrow x^2 + 2x - 24 = 0$$

$$\Rightarrow (x + 6)(x - 4) = 0 \Rightarrow x = 4 \text{ only}$$

$$\text{case (ii) } x \in (-3, 3) \Rightarrow x^2 = 18 \Rightarrow \text{No sol.}$$

$$\text{case (iii) } x < -3 \Rightarrow x^2 - 2x - 24 = 0$$

$$\Rightarrow (x - 6)(x + 4) = 0 \Rightarrow x = -4 \text{ only}$$

$$\therefore x = \pm 4 \Rightarrow \text{sum of solutions} = 0$$

9. **Ans (4)**

$$g \circ f(x) = 3x + 4 ; -1 \leq x \leq 0$$

$$\text{Range} = [1, 4]$$



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ANSWER KEY

PAPER-2

PHYSICS

Section-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	B	D	D	A	D	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
Section-II	A.	D	B	D	C	C	B	C	B	D	C
	Q.	1	2	3	4	5	6	7	8	9	10
Section-II	A.	45.00	1.00	0.20	0.25	0.50	1130.00	1.50	15.00	0.03 to 0.04	3.00

CHEMISTRY

Section-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	C	D	B	B	B	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
Section-II	A.	C	D	D	A	A	D	B	B	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
Section-II	A.	1.00	4.00	13.88 to 13.89	81.00	2.00	5.00	3.00	2.00	4.00	7.00

MATHEMATICS

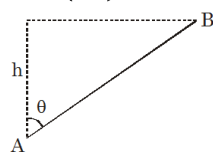
Section - A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	B	C	B	B	B	A	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
Section-II	A.	C	B	A	C	B	D	A	B	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
Section-II	A.	11.00	51.00	35.00	20.00	2.00	0.00	5.00	3.00	2.00	25.00

HINT – SHEET

PHYSICS

Section-I

1. Ans (B)



$$F_H = \rho \cdot g \frac{h}{2} \cdot h \cdot b$$

$$= \frac{1}{2} \rho \cdot g h^2 b$$

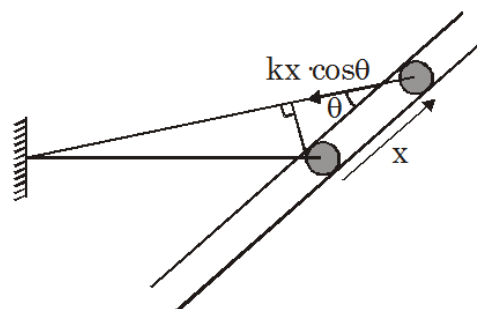
$$F_V = \frac{1}{2} \cdot h \cdot h \tan \theta \cdot b \cdot \rho \cdot g$$

$$= \frac{1}{2} \rho \cdot g \cdot h^2 b \tan \theta$$

$$F = \sqrt{F_H^2 + F_V^2}$$

$$F = \frac{1}{2} \rho \cdot g h^2 b \cdot \sec \theta$$

2. Ans (A)



$$-(kx \cdot \cos \theta) \cdot \cos \theta = ma$$

$$a = - \left(\frac{k \cos^2 \theta}{m} \right) x$$

$$\omega = \cos \theta \sqrt{\frac{k}{m}}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\cos \theta} \cdot \sqrt{\frac{m}{k}}$$

4. **Ans (B)**

$$F = \rho A$$

$$= \rho \cdot g \cdot 3l \cdot A$$

$$= 3\rho \cdot g l A$$

5. **Ans (D)**

$$U = U_0 [1 - \cos(bx)]$$

$$F = -\frac{dU}{dx} = -U_0 b \cdot \sin(bx)$$

For small oscillation

$$F = -U_0 b (bx)$$

$$a = -\left(\frac{U_0 b^2}{m}\right)x$$

$$T = \frac{2\pi}{b} \sqrt{\frac{m}{U_0}}$$

6. **Ans (D)**

$$E = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} m \cdot \frac{k}{m} \cdot A^2 = \frac{1}{2} k A^2$$

$$E' = \frac{1}{2} k A^2$$

$$E' = E$$

7. **Ans (A)**

$$P_0 + \rho g H - \frac{\rho g}{4} \cdot L = P_0$$

$$L = 4H$$

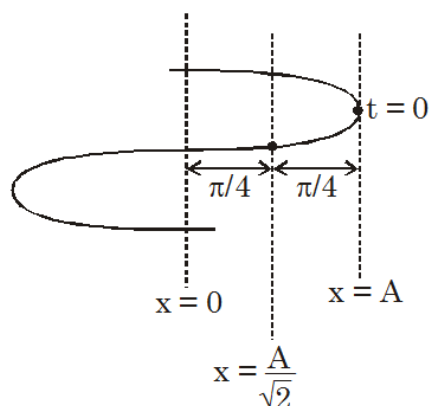
8. **Ans (D)**

$$T = \frac{1}{2} \cdot 2\pi \sqrt{\frac{m}{k}} + \frac{1}{2} \cdot 2\pi \sqrt{\frac{m}{2k}}$$

$$= \pi \sqrt{\frac{m}{k}} + \pi \sqrt{\frac{m}{2k}}$$

$$= \pi \sqrt{\frac{m}{k}} \left(1 + \frac{1}{\sqrt{2}}\right)$$

9. **Ans (B)**



$$\Delta \phi = \omega \cdot \Delta t$$

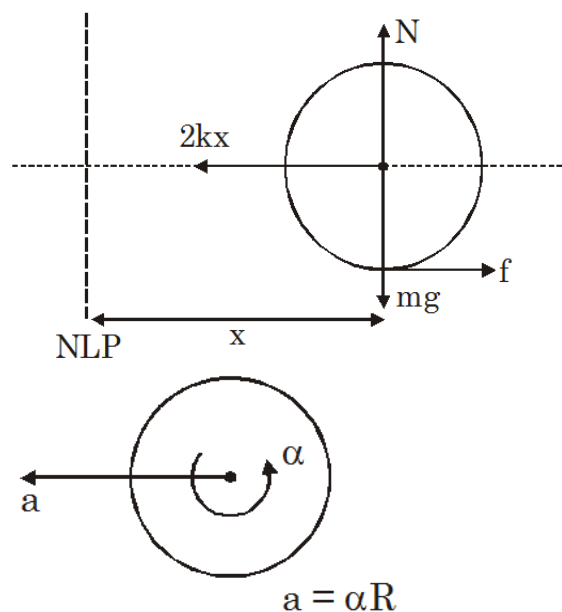
$$= \omega \cdot \frac{\pi}{4\omega}$$

$$= \frac{\pi}{4}$$

$$a = -\omega^2 x$$

$$= -\omega^2 \cdot \frac{A}{\sqrt{2}}$$

10. **Ans (A)**



$$2kx - f = ma \quad \dots (i)$$

$$fR = \frac{2}{5} m R^2 \cdot \alpha$$

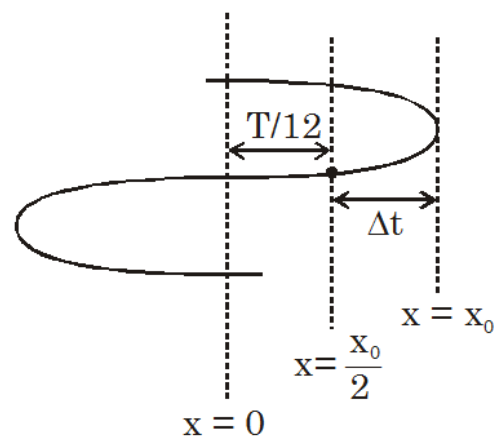
$$f = \frac{2}{5} ma \quad \dots (ii)$$

$$2kx = \frac{7}{5} ma$$

$$a = \frac{10}{7} \frac{kx}{m}$$

$$\omega = \sqrt{\frac{10k}{7m}}$$

$$T = 2\pi \sqrt{\frac{7m}{10k}}$$



$$\Delta t = \frac{T}{4} - \frac{T}{12} = \frac{T}{6}$$

$$= \frac{\pi}{3} \sqrt{\frac{7m}{10k}}$$

11. Ans (D)

$$g_{\text{eff}} = \sqrt{g^2 + \left(\frac{v^2}{R}\right)^2} = \sqrt{g^2 + g^2} = g\sqrt{2}$$

$$T = 2\pi\sqrt{\frac{\ell}{g_{\text{eff}}}}$$

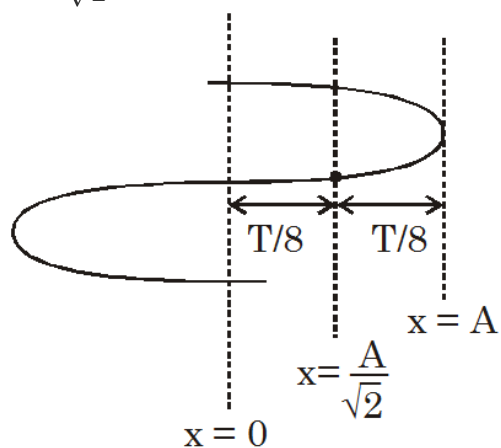
$$= 2\pi\sqrt{\frac{\ell}{g\sqrt{2}}}$$

12. Ans (B)

$$v = \omega\sqrt{A^2 - x^2}$$

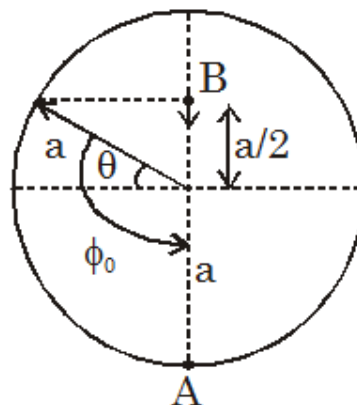
$$\frac{A\omega}{\sqrt{2}} = \omega\sqrt{A^2 - x^2}$$

$$x = \frac{A}{\sqrt{2}}$$



$$\Delta t = \frac{T}{8} = \frac{2\pi}{8} \sqrt{\frac{m}{k}}$$

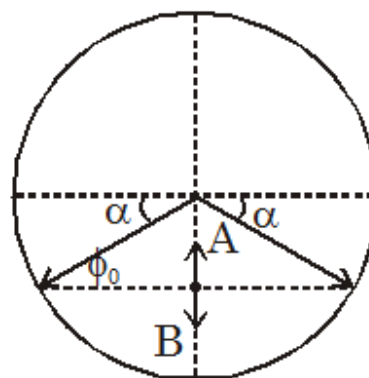
13. Ans (D)



$$\phi_0 = \theta + \pi/2$$

$$\theta = \pi/6$$

$$\phi_0 = \pi/6 + \pi/2 = \frac{4\pi}{6} = \frac{2\pi}{3}$$



$$\theta + \alpha = \omega\Delta t$$

$$\frac{\pi}{2} - \alpha = \omega\Delta t$$

$$\theta + \frac{\pi}{2} = 2\omega\Delta t$$

$$\Delta t = \frac{\theta + \frac{\pi}{2}}{2\omega} = \frac{\frac{2\pi}{3}}{2\omega} = \frac{\pi}{3\omega}$$

14. Ans (C)

$$8 = \frac{1}{2} kA^2$$

$$A = 6 \text{ cm}$$

$$k = \frac{16}{(6 \times 10^{-2})^2} = \frac{16}{36} \times 10^4$$

$$= \frac{40}{9} \times 10^3 = 4.44 \text{ kN/m}$$

mass can't be found k_{max} at mean.

15. **Ans (C)**

$$a_{\max} \text{ of } m = \frac{1}{m} (\mu mg \cos \theta - mg \sin \theta) = g \sin \theta$$

$$\Rightarrow (kx - 3mg \sin \theta) = 3m \cdot a_{\max}$$

$$\Rightarrow x = \frac{6mg \sin \theta}{k}$$

$$\text{Amplitude} = x - \frac{3mg \sin \theta}{k} = \frac{3mg \sin \theta}{k}$$

17. **Ans (C)**

$$I_{\text{CM}} = mk^2$$

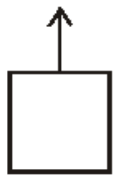
$$\frac{mL^2}{12} = mk^2$$

$$k = \frac{L}{\sqrt{12}}$$

$$\text{Distance} = \frac{L}{2} + \frac{L}{\sqrt{12}}$$

18. **Ans (B)**

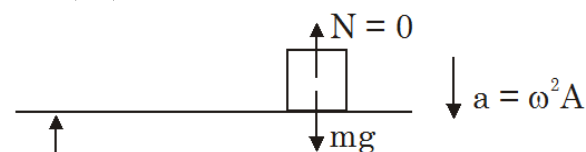
$$F_{\text{net}} = \rho A x g$$



$$F_{\text{net}} = \rho A x g \Rightarrow \frac{\rho}{3} A \ell g \times a = \rho A x g \Rightarrow a = \frac{3x}{\ell}$$

$$\text{time period} = 2\pi \sqrt{\frac{\ell}{3g}}$$

19. **Ans (D)**



$$mg = m\omega^2 A$$

$$g = \frac{k}{(M+m)} A$$

$$A = \frac{(M+m)g}{k}$$

20. **Ans (C)**

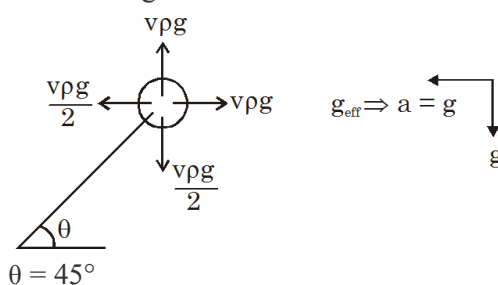
No effect on length inside the liquid, due to motion of container in vertical direction, having constant acceleration.

PHYSICS

Section-II

1. **Ans (45.00)**

$$\tan 45^\circ = \frac{a}{g} \Rightarrow a = g$$



2. **Ans (1.00)**

$$\rho gh = \frac{1}{2} \rho v^2$$

$$v = \sqrt{2gh}$$

$$= \sqrt{2 \times 10 \times \frac{5}{100}}$$

$$= 1 \text{ m/s}$$

3. **Ans (0.20)**

$$t \times v_{\max} = \frac{2\pi}{\omega} \cdot A\omega = 2\pi A$$

$$= 2\pi \times \frac{10}{\pi} \text{ cm}$$

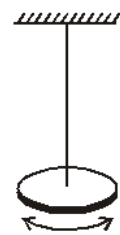
$$= 0.20 \text{ m}$$

4. **Ans (0.25)**

The situation is shown in figure. the moment of inertia of the disc about the wire is

$$I = \frac{mr^2}{2} = \frac{(0.200 \text{ kg})(5.0 \times 10^{-2} \text{ m})^2}{2}$$

$$= 2.5 \times 10^{-4} \text{ kg-m}^2.$$



The time period is given by

$$T = 2\pi \sqrt{\frac{I}{k}}$$

$$\text{or, } k = \frac{4\pi^2 I}{T^2}$$

$$= \frac{4\pi^2 (2.5 \times 10^{-4} \text{ kg-m}^2)}{(0.20 \text{ s})^2}$$

$$= 0.25 \frac{\text{kg-m}^2}{\text{s}^2}.$$

5. **Ans (0.50)**

Soap solution film will be formed on both sides of the frame. Each film is in contact with the wire along a distance of 10 cm. The force exerted by the film on the wire

$$= 2 \times (10 \text{ cm}) \times (25 \text{ dyne/cm})$$

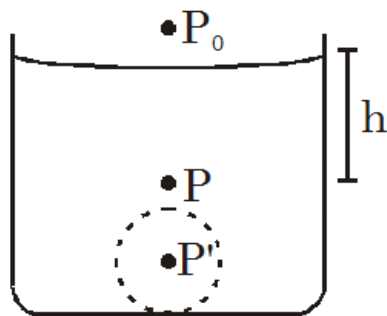
$$= 500 \text{ dyne} = 5 \times 10^{-3} \text{ N.}$$

This force acts vertically upward and should be balanced by the load. Hence the load that should be suspended is

$5 \times 10^{-3} \text{ N}$. The mass of the load should be

$$\frac{5 \times 10^{-3} \text{ N}}{10 \text{ m/s}^2} = 5 \times 10^{-4} \text{ kg} = 0.5 \text{ g.}$$

6. **Ans (1130.00)**



Let the atmospheric pressure be P_0 . The pressure of the liquid just outside the bubble is (figure)

$$P = P_0 + h\rho g.$$

The pressure inside the bubble is

$$P' = P + \frac{2S}{r} = P_0 + h\rho g + \frac{2S}{r}$$

or,

$$P' - P_0 = (10 \text{ cm}) (1000 \text{ kg/m}^3) (9.8 \text{ m/s}^2) +$$

$$\frac{2 \times 0.075 \text{ N/m}}{1.0 \times 10^{-3} \text{ m}}$$

$$= 980 \text{ N/m}^2 + 150 \text{ N/m}^2$$

$$= 1130 \text{ Pa.}$$

7. **Ans (1.50)**

$$P_0 - \frac{2s}{r} + \rho gh + \frac{2s}{r} = P_0$$

$$\frac{2s}{r} = \frac{2s}{r} - \rho gh$$

$$\frac{1}{r} = \frac{1}{r} - \frac{\rho gh}{2s}$$

$$= \frac{1}{0.5 \times 10^{-3}} - \frac{10^3 \times 10 \times 0.02}{2 \times 0.075}$$

$$r' = 1.5 \text{ mm}$$

8. **Ans (15.00)**

$$\text{For a simple pendulum } T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}}$$

situation 1 : when pendulum is in air

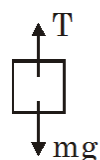
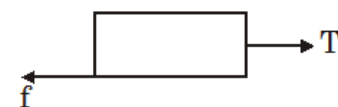
$$\rightarrow g_{\text{eff}} = g$$

situation 2 : when pendulum is in liquid

$$\rightarrow g_{\text{eff}} = g \left(1 - \frac{\rho_{\text{liquid}}}{\rho_{\text{body}}} \right) = g \left(1 - \frac{1}{16} \right) = \frac{15g}{16}$$

$$\text{So, } \frac{T'}{T} = \frac{2\pi \sqrt{\frac{L}{15g/16}}}{2\pi \sqrt{\frac{L}{g}}} \Rightarrow T' = \frac{4T}{\sqrt{15}}$$

9. **Ans (0.03 to 0.04)**



$$T = mg$$

$$T = f$$

$$mg = \eta \cdot A \frac{\Delta v}{\Delta y}$$

$$\eta = \frac{mg \cdot \Delta y}{A \cdot \Delta v}$$

$$= \frac{0.01 \times 9.8 \times 3 \times 10^{-3}}{0.1 \times 0.085}$$

$$= 0.03458$$

10. **Ans (3.00)**

$$\omega = \sqrt{\frac{k}{I}} = \sqrt{\frac{12k}{7mL^2}}$$

$$v = \theta_0 \cdot \omega \cdot \frac{L}{2}$$

$$= \theta_0 \frac{L}{2} \sqrt{\frac{12k}{7mL^2}}$$

$$v = \theta_0 \sqrt{\frac{3k}{7m}}$$

$$\alpha = 3$$

CHEMISTRY

Section-I

2. **Ans (B)**

$$\text{Volume of 4 spheres} = 4 \times \frac{4}{3}\pi r^3$$

$$\text{Volume of cuboid} = \frac{(4r)^2 \times 2r}{\text{Area} \times \text{height}}$$

Percentage of vacant space =

$$\frac{(4r)^2 \times 2r - 4 \times \frac{4}{3}\pi r^3}{(4r)^2 \times 2r} \times 100 = \left(1 - \frac{\pi}{6}\right) \times 100$$

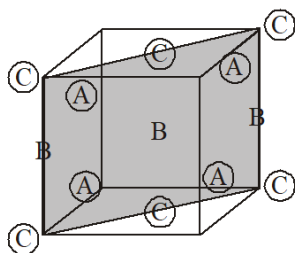
3. **Ans (D)**

$$\text{Edge length} = 2(r_{\text{Na}^+} + r_{\text{Cl}^-}) = a$$

Shorted distance between Na⁺ ions will be along face diagonal

$$= \frac{\sqrt{2}a}{2} = \frac{a}{\sqrt{2}} = \frac{2(95 + 181)}{\sqrt{2}} = 390.3 \text{ pm}$$

4. **Ans (C)**



C = Corner + Face centres

B = Edge centre (octahedral void) + Body centre (octahedral void)

A = Tetrahedral void



10. **Ans (B)**

Each iodine is sp³d² hybridised & chlorine is sp³ hybridised.

18. **Ans (B)**

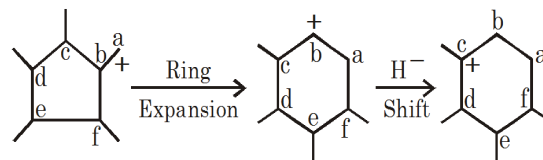
Sulphur is strongest nucleophile due to size. Strong base are good nucleophile (if charge is present on same type of atom)

19. **Ans (D)**

Leaving group ability $\propto$ its stability

Weak base are good leaving group

20. **Ans (D)**



CHEMISTRY

Section-II

2. **Ans (4.00)**

$$\frac{r_+}{r_-} = \frac{1.6}{1.864} = 0.858$$

So, it is CsCl type unit cell

$$\text{So } \sqrt{3}a = 2(r_+ + r_-)$$

$$\text{So } a = \frac{2(1.864 + 1.6)}{\sqrt{3}} \text{ \AA} = 2 \times 2 \text{ \AA} = 4 \text{ \AA}$$

5. **Ans (2.00)**

SN

MATHEMATICS

Section - A

1. **Ans (B)**

$$\alpha + \beta = \frac{-b}{a}, \quad \alpha\beta = \frac{c}{a}$$

$$(1 + \alpha + \alpha^2)(1 + \beta + \beta^2)$$

$$= 1 + \beta + \beta^2 + \alpha + \alpha\beta + \alpha\beta^2 + \alpha^2 + \alpha^2\beta + \alpha^2\beta^2$$

$$= 1 + (\alpha + \beta) + (\alpha^2 + \beta^2) + \alpha\beta(\alpha + \beta) + \alpha\beta + \alpha^2\beta^2$$

$$= 1 + \left(\frac{-b}{a}\right) + (\alpha + \beta)^2 - \alpha\beta + \alpha\beta(\alpha + \beta) + \alpha^2\beta^2$$

$$= 1 - \frac{b}{a} + \frac{b^2}{a^2} - \frac{c}{a} - \frac{b}{a} \frac{c}{a} + \frac{c^2}{a^2}$$

$$= 1 - \frac{(b+c)}{a} + \frac{b^2+c^2}{a^2} - \frac{bc}{a^2}$$

$$= \frac{a^2 - ab - ac + b^2 + c^2 - bc}{a^2}$$

$$= \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2a^2} > 0$$

2. **Ans (B)**

$$f(x) = ax^2 + bx + c$$

Case-1 $a > 0$

$$f(1) = a + b + c > 0$$

$$f(0) = c > 0$$

Case-2 $a < 0$

$$f(1) = a + b + c < 0$$

$$f(0) = c < 0$$

in both case

$$c(a + b + c) > 0$$

3. **Ans (B)**

$$(x + 2)^2 + (x^2 + 5x + 6)^2 + (x^2 + 3x + 2)^2 = 0$$

possible only if

$$x + 2 = 0 \text{ and } x^2 + 5x + 6 = 0 \text{ and } x^2 + 3x + 2 = 0$$

$$\Rightarrow x = -2 \text{ and } x = -2, -3 \text{ and } x = -1, -2$$

So $x = -2$ only one value of x .

4. **Ans (C)**

$$x^2 + 2ax + a^2 - 1 = 0$$

$$(x + a)^2 = 1$$

$$x + a = \pm 1$$

$$x = 1 - a, -1 - a$$

at least one root is negative so

$$-a - 1 < 0 \Rightarrow a > -1$$

$$a \in (-1, \infty)$$

5. **Ans (B)**

$$\text{Here } 0 \leq \{x\} < 1$$

$$\Rightarrow 0 \leq \sin\{x\} < \sin 1$$

But functions $f(x)$ defined so $\sin\{x\} \neq 0$

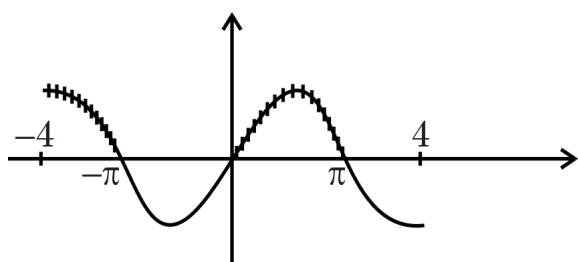
$$\Rightarrow \frac{1}{\sin\{x\}} \in \left(\frac{1}{\sin 1}, \infty\right)$$

$$\Rightarrow \left[\frac{1}{\sin\{x\}}\right] \in \{1, 2, 3, \dots\}$$

6. **Ans (B)**

$$\sin x \geq 0 \text{ and } 16 - x^2 > 0$$

$$x \in [-4, 4]$$



$$x \in [-4, -\pi] \cup [0, \pi]$$

7. **Ans (B)**

$$f(x) = \left(\sin x - \frac{1}{2}\right)^2 - \frac{5}{4}$$

$$\text{Range} \left[-\frac{5}{4}, 1\right]$$

8. **Ans (A)**

$$y = (x - 1)^2 + 2$$

$$\text{Range} = [2, 6]$$

9. **Ans (D)**

$$\frac{x}{x^2} = \frac{1}{x} \quad x \neq 0$$

equal

10. **Ans (D)**

$$e^x \in (0, 1)$$

$$x \in (-\infty, 0)$$

11. **Ans (C)**

$$f(x + \pi/2) = \cos^2 x + \sin^4 x$$

$$= \cos^2 x + (1 - \cos^2 x)^2$$

$$= \cos^2 x + \cos^4 x - 2\cos^2 x + 1$$

$$= \cos^4 x + \sin^2 x = f(x)$$

12. **Ans (B)**

$$f(x) = \frac{\pi}{2} + \tan^{-1} x \quad x \in [-1, 1]$$

$$\left[\frac{\pi}{2} + \left(-\frac{\pi}{4}\right), \frac{\pi}{2} + \left(\frac{\pi}{4}\right)\right]$$

$$\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

13. **Ans (A)**

$$4 \left[\pi - \cos^{-1}(\sin 5) + 3 \tan^{-1} \left(\cot \left(\frac{1}{4} \right) \right) \right]$$

$$= 4 \left[\pi - \frac{\pi}{2} + \sin^{-1} \sin 5 + \frac{3\pi}{2} - \frac{3}{4} \right]$$

$$= 4 \left[\frac{\pi}{2} + 5 - 2\pi + \frac{3\pi}{2} - \frac{3}{4} \right]$$

$$= 4 \left[\frac{17}{4} \right] = 17$$

14. **Ans (C)**

$$-1 \leq 1 - x \leq 1 \text{ \& } -1 \leq x \leq 1$$

$$x \in [0, 1]$$

$$\cos^{-1}(1 - x) = \frac{\pi}{2} + 3\cos^{-1} x$$

$$\text{LHS} \in \left[0, \frac{\pi}{2}\right], \text{ RHS} \in \left[\frac{\pi}{2}, 2\pi\right]$$

possible only when $x = 1$

15. Ans (B)

$$\begin{aligned}\tan^{-1} \left(\frac{1}{1+r+r^2} \right) &= \tan^{-1} \left(\frac{(r+1)-(r)}{1+r(r+1)} \right) \\ &= \tan^{-1}(r+1) - \tan^{-1}(r) \\ S_n &= \tan^{-1}(2) - \tan^{-1}(1) + \\ &\quad \tan^{-1}(3) - \tan^{-1}(2) + \\ &\quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\ &\quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \\ &\quad \tan^{-1}(n+1) - \tan^{-1}(n) \\ S_n &= \tan^{-1}(n+1) - \frac{\pi}{4} \\ S_\infty &= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}\end{aligned}$$

16. Ans (D)

$$\begin{aligned}\text{Let } \theta &= \tan^{-1}x, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \\ \cos^{-1}(\cos 2\theta) &\leq \frac{\pi}{3}, 2\theta \in (-\pi, \pi) \\ -2\theta &\leq \frac{\pi}{3} \text{ \& } -\pi < 2\theta \leq \pi \\ \Rightarrow -\frac{\pi}{6} &\leq \theta \leq 0 \\ \Rightarrow -\frac{1}{\sqrt{3}} &\leq x \leq 0 \\ \text{or } 2\theta &\leq \frac{\pi}{3} \text{ \& } 0 < 2\theta < \pi \\ 0 < \theta &\leq \frac{\pi}{6} \Rightarrow x \in \left(0, \frac{1}{\sqrt{3}} \right] \\ x \in \left[\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right], &x = 0 \text{ is the only integral} \\ \text{value}\end{aligned}$$

17. Ans (A)

$$\begin{aligned}1 \leq |x-1| - 2 < 2 \text{ or } -2 < |x-1| - 2 \leq -1 \\ 3 \leq |x-1| < 4 \text{ or } 0 < |x-1| \leq 1 \\ 3 \leq x-1 < 4 \text{ or } -4 < x-1 \leq -3 \text{ or } \\ -1 \leq x-1 < 0 \text{ or } 0 < x-1 \leq 1 \\ 4 \leq x < 5 \text{ or } -3 < x \leq -2 \text{ or } \\ 0 \leq x < 1 \text{ or } 1 < x \leq 2 \\ \therefore x \in (-3, -2] \cup [0, 1) \cup (1, 2] \cup [4, 5) \\ \therefore x^2 \in [4, 9) \cup [0, 1) \cup (1, 4] \cup [16, 25) \\ x^2 \in [0, 1) \cup (1, 9) \cup [16, 25)\end{aligned}$$

18. Ans (B)

$$\begin{aligned}\text{As } p(x) \text{ is odd function} \\ \text{Hence } P(-x) = -P(x) \Rightarrow P(-3) = -P(3) = -6 \\ \text{Let } P(x) = Q(x^2 - 9) + ax + b \quad \dots(1) \\ P(-3) = -3a + b = -6 \quad \dots(2) \\ \Rightarrow b = 0 \text{ and } a = 2 \\ \Rightarrow g(x) = 2x \\ \Rightarrow g(2) = 4\end{aligned}$$

19. Ans (C)

$$\begin{aligned}\because f(x) &= (x^2 - 1)(x^2 - 4)(x^2 - 9) + x^2 + 4 \\ \therefore f(\sqrt{2}) &= 20 \text{ and } f(\sqrt{3}) = 19 \\ f(\sqrt{2}) - f(\sqrt{3}) &= 1\end{aligned}$$

20. Ans (D)

$$\begin{aligned}\because |x^2 - x + 2 + f(x)| &= |x^2 - x + 2| + |f(x)| \quad \forall x \in \mathbb{R} \\ \Rightarrow (x^2 - x + 2) f(x) &\geq 0 \quad \forall x \in \mathbb{R} \\ \Rightarrow f(x) &\geq 0 \quad \forall x \in \mathbb{R} \text{ but } f(1) = 0 \\ \Rightarrow \therefore f(x) &= (x - 1)^2 \text{ (as } f(x) \text{ is monic polynomial)} \\ \therefore f(3) &= 4\end{aligned}$$

MATHEMATICS

Section-II

1. Ans (11.00)

$$\begin{aligned}\text{Let } y &= \frac{x-2}{x^2-x+2} \\ \Rightarrow yx^2 - yx + 2y - x + 2 &= 0 \\ \Rightarrow yx^2 - x(1+y) + 2y + 2 &= 0 \\ x \in \mathbb{R} \Rightarrow D &> 0 \\ (y+1)^2 - 4y(2y+2) &\geq 0 \Rightarrow (y+1)(y+1-8y) \\ &\geq 0 \\ (y+1)(7y-1) &\leq 0 \\ y \in \left[-1, \frac{1}{7} \right] \Rightarrow \alpha = -1, \beta = \frac{1}{7} \\ 10|\alpha| + 7|\beta| &= 10|-1| + 7\left|\frac{1}{7}\right| = 11\end{aligned}$$

2. **Ans (51.00)**

If $[x] = 0$, then $[y] = 50$ and the area enclosed = 1 unit

If $[x] = 1$, then $[y] = 49$ and the area enclosed = 1 unit

If $[x] = 2$, then $[y] = 48$ and the area enclosed = 1 unit

If $[x] = 50$, then $[y] = 0$ and the area enclosed = 1 unit

$\therefore$ the total enclosed area = 51 units

3. **Ans (35.00)**

$$-x^2 - 2x + n \geq 0$$

downward parabola (+)ve between roots

$$\text{roots, } x = -1 \pm \sqrt{1+n}$$

$$\text{thus, } \Rightarrow -1 \pm \sqrt{1+n} \geq 5 \Rightarrow n \geq 35$$

4. **Ans (20.00)**

$$f(x) = \begin{cases} x & x \in \mathbb{Q} \\ 1-x & x \notin \mathbb{Q} \end{cases}$$

;

$$f \circ f(x) = \begin{cases} f(x) & f(x) \in \mathbb{Q} \\ 1-f(x) & f(x) \notin \mathbb{Q} \end{cases}$$

$$= \begin{cases} x & x \in \mathbb{Q} \\ 1-(1-x) & 1-x \notin \mathbb{Q} \end{cases} = \begin{cases} x & x \in \mathbb{Q} \\ x & x \notin \mathbb{Q} \end{cases}$$

$$\therefore f(f(x)) = x; a = 0; b = 1$$

5. **Ans (2.00)**

Here $f(x) = [x]$ and $g(x) = |x|$

$$(f \circ g)x = f(g(x)) = f(|x|) = [|x|]$$

$$(f \circ g)\left(\frac{-3}{2}\right) = \left[\left|\frac{-3}{2}\right|\right] = \left[\frac{3}{2}\right] = 1$$

$$(g \circ f)x = g(f(x)) = g([x]) = |[x]|$$

$$(g \circ f)\left(\frac{4}{3}\right) = \left|\left[\frac{4}{3}\right]\right| = |1| = 1$$

$$(f \circ g)\left(\frac{-3}{2}\right) + (g \circ f)\left(\frac{4}{3}\right) = 1 + 1 = 2$$

6. **Ans (0.00)**

$$\sin(4\theta) = 2\sin 2\theta \cos 2\theta; \theta = \tan^{-1} \frac{1}{3}$$

$$\therefore 2\theta = 2\tan^{-1} \frac{1}{3} = \tan^{-1} \left(\frac{3}{4}\right)$$

$$= 2\sin\left(\tan^{-1} \frac{3}{4}\right) \cos\left(\tan^{-1} \frac{3}{4}\right)$$

$$= 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$$

$$\therefore \cos\left(2\tan^{-1} \frac{1}{3}\right) = \cos\left(\tan^{-1} \frac{7}{24}\right) = \frac{24}{25}$$

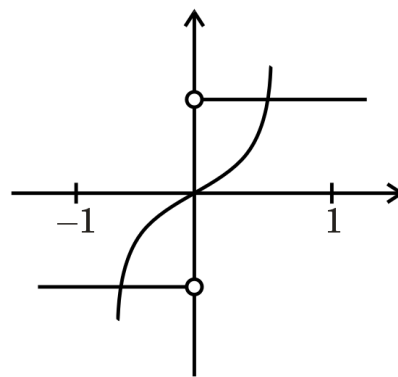
7. **Ans (5.00)**

$$\therefore f(x) = \cot^{-1}(\cot x)$$

$$\therefore \sum_{x=1}^5 f(x) = \cot^{-1}(\cot 1) + \cot^{-1}(\cot 2) + \cot^{-1}(\cot 3) + \cot^{-1}(\cot 4) + \cot^{-1}(\cot 5)$$

$$= 1 + 2 + 3 + 4 - \pi + 5 - \pi = 15 - 2\pi$$

8. **Ans (3.00)**



9. **Ans (2.00)**

$$D \geq 0$$

$$(2a+b)^2 - 4(2a^2+b^2-b+\frac{1}{2}) \geq 0$$

$$4a^2 + 3b^2 - 4ab - 4b + 2 \leq 0$$

$$(2a-b)^2 + 2(b-1)^2 \leq 0$$

which is only possible when $b = 1$, $2a = b$ i.e. $a = \frac{1}{2}$

$$= \frac{1}{2}$$

10. **Ans (25.00)**

$$\text{Given } \frac{2x}{2x^2 + 5x + 2} > \frac{1}{x+1}$$

$$\Rightarrow \frac{2x}{(2x+1)(x+2)} > \frac{1}{(x+1)}$$

$$\Rightarrow \frac{2x}{(2x+1)(x+2)} - \frac{1}{(x+1)} > 0$$

$$\Rightarrow \frac{2x(x+1) - (2x+1)(x+2)}{(x+1)(x+2)(2x+1)} > 0$$

$$\Rightarrow \frac{3x+2}{(x+1)(x+2)(2x+1)} < 0$$

$$\begin{array}{ccccccc} + & - & + & - & + \\ -2 & -1 & -2/3 & -1/2 & \end{array}$$

$$\therefore x \in (-2, -1) \cup \left(-\frac{2}{3}, -\frac{1}{2}\right)$$